

Let the function $\mu_r(t)$ be:

$$\mu_r(t) = \int_0^t \lambda_r(v) dv = (1 + r \frac{\gamma}{\rho}) \ln(1 + \rho t). \quad (\text{A.3})$$

It can be seen from Eq. (A.3) that $\mu'_r(t) = \lambda_r(t)$ and $\mu_r(0) = 0 \quad \forall r \geq 0$.

We now define the auxiliary function

$$\mu_\Delta(t) = \mu_{r+1}(t) - \mu_r(t) = \frac{\gamma}{\rho} \ln(1 + \rho t), \quad (\text{A.4})$$

which does not depend on r . Note that $\mu_\Delta(0) = 0$.

It is a fact that the following equality holds:

$$\int_0^t \lambda_r(x) e^{\mu_\Delta(x)} (e^{\mu_\Delta(x)} - 1)^r dx = \frac{(\frac{\rho}{\gamma} + r)}{r + 1} (e^{\mu_\Delta(t)} - 1)^{r+1}. \quad (\text{A.5})$$

The proof can be done by differentiating the right side of Eq. (A.5) to show it's indeed a primitive of the left side's integrand. Since the functions are also equal on $t = 0$, fundamental theorem of calculus yields that the equality is valid for every t .

Considering the initial condition $P_0(0) = 1$ (the beginning population is 0), the probability mass functions are given by:

$$P_r(t) = \frac{\Gamma(\frac{\rho}{\gamma} + r)}{r! \Gamma(\frac{\rho}{\gamma})} e^{-(\mu_r(t) - \mu_0(0))} (e^{\mu_\Delta(t)} - 1)^r. \quad (\text{A.6})$$

This is demonstrated by induction over r , using Eq. (A.6) as inductive hypothesis and $P_0(t)$ from Eq. (A.1). In fact,

$$\begin{aligned} P_{r+1}(t) &= \int_0^t e^{-(\mu_{r+1}(t) - \mu_{r+1}(x))} \lambda_r(x) P_r(x) dx \\ &= \frac{\Gamma(\frac{\rho}{\gamma} + r)}{r! \Gamma(\frac{\rho}{\gamma})} e^{-(\mu_{r+1}(t) - \mu_0(0))} \int_0^t \lambda_r(x) e^{\mu_\Delta(x)} (e^{\mu_\Delta(x)} - 1)^r dx \\ &= \frac{\Gamma(\frac{\rho}{\gamma} + r)}{r! \Gamma(\frac{\rho}{\gamma})} e^{-(\mu_{r+1}(t) - \mu_0(0))} \frac{(\frac{\rho}{\gamma} + r)}{r + 1} (e^{\mu_\Delta(t)} - 1)^{r+1}. \end{aligned}$$

We can rewrite Eq. (A.6) as:

$$P_r(t) = \frac{\Gamma(\frac{\rho}{\gamma} + r)}{r! \Gamma(\frac{\rho}{\gamma})} e^{-(\mu_r(t) - \mu_0(0) - r\mu_\Delta(t))} (1 - e^{-\mu_\Delta(t)})^r. \quad (\text{A.7})$$